



F1-12-MKII Hydrostatic Pressure

The Hydrostatic Pressure accessory has been designed to determine the static thrust exerted by a fluid on a submerged surface and enables comparison of the measured magnitude and position of this force with simple theory. A fabricated quadrant is mounted on a balance arm, which pivots on knife edges. The knife edges coincide with the centre of the arc of the quadrant. This means that when the quadrant is immersed, the only force that gives rise to a moment about the knife edges is the hydrostatic force acting on the end face of the quadrant.

PRODUCT CODE: F1-12-MKII

Categories: Agricultural, Educational, F1 Series Fluid Mechanics, F1-12-MKII Hydrostatic Pressure

Tags: Armfield , Determine Static Thrust. , F1-12 , Fluid Mechanics , Hydrostatic Pressure , Knife Edge Piviot , Submerged Plane , Submerged Surface , Theoretical Position

DESCRIPTION

The Hydrostatic Pressure accessory has been designed to determine the static thrust exerted by a fluid on a submerged surface and enables comparison of the measured magnitude and position of this force with simple theory.

A fabricated quadrant is mounted on a balance arm which pivots on knife edges. The knife edges coincide with the centre of the arc of the quadrant. This means that when the quadrant is immersed, the only force that gives rise to a moment about the knife edges is the hydrostatic force acting on the end face of the quadrant.

The balance arm incorporates a hanger for the weights supplied and an adjustable counterbalance.

This assembly is mounted on top of an acrylic tank which may be levelled by adjusting screwed feet. Correct alignment is indicated on a circular spirit level mounted on the base of the tank. An indicator attached to the side of the tank shows when the balance arm is horizontal. Water is added to the tank via a supplied jug and may be drained through a valve in the side of the tank. The water level is indicated on a scale on the side of the quadrant.

TECHNICAL SPECIFICATIONS

Tank capacity: 5.5l

Distance between suspended mass and fulcrum: 0.266m

Cross-sectional area of quadrant (toroid): $7.5 \times 10^{-3} \text{m}^2$
Total depth of completely immersed quadrant: 170mm
Height of fulcrum above quadrant: 100mm

FEATURES & BENEFITS

Experimental Content

- To determine the hydrostatic thrust acting on a plane surface immersed in water when the surface is partially submerged or fully submerged
- To determine the position of the line of action of the thrust and to compare the position determined by experiment with the theoretical position

Practical Training Exercises

Graph plotting r against mass using F1-12 (indicative)

F1-12-MKII Videos (English - Español)



Dimensions

Length: 0.417m

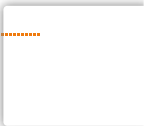
Width: 0.230m

Height: 0.302m

Order Codes

F1-12-MKII

YOU MAY ALSO LIKE...





**F1-14 MKII
METACENTRIC
HEIGHT**

[Read more](#)



**F1-10
HYDRAULICS
BENCH**

[Read more](#)



**F9092 – FLUID
PROPERTIES &
HYDROSTATICS
BENCH**

[Read more](#)

Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781

